

Flowsync: Period and Hormonal Health Tracker

Abstract/ Executive Summary

FlowSync focuses on the question of how menstrual health and hormonal health tracking on an app can be improved by using personalized data especially for women with conditions like PCOS, PMDD, and PMS. The importance of this product is to provide a period tracker that goes hand in hand with hormonal imbalance which allow women to prepare early for the symptoms or detect underlying health conditions. User research, diverse AI model training data, and wearable component (smart watch) is needed for product development. The expected outcome for this mobile app is to provide an inclusive experience for any woman.

Introduction

Currently in the market, most period tracker apps don't adapt to users' irregular cycles or hormonal imbalances. This concept allows menstrual and hormonal health tracking systems to improve through the use of predictive analytics and personalized data. A study has shown that while period tracking accuracy increases, apps like Clue and Natural Cycles are still not personalized. FlowSync can be impactful to women whose needs are not met by existing apps. This app can make correct predictions for women with irregular cycles, detect early underlying health issues, and provide tailored lifestyle recommendations such as diet, rest times, and exercise for all women.

Research steps needed to answer the question at hand

Quantitative and qualitative data are essential for accurate, personalized results in this app. To ensure inclusivity, women with PMDD, PCOS, and PMS are represented through user surveys and interviews. Human-centered design will guide development via iterative prototyping and continuous feedback. Physiological data like temperature, heart rate variability (HRV), and skin conductance can detect ovulation, predict periods, identify irregularities, and highlight stress or hormonal imbalances. For example, body temperature can rise post-ovulation¹. HRV can drop before menstruation which can indicate stress or imbalance². Skin conductance tracks subtle sweat changes tied to stress³. Time series data can predict cycle and detect irregularities, and deep learning models can understand complex hormonal patterns^{4,5}. This device can be tested women with PMDD, PCOS, or other underrepresented conditions which helps evaluate its accuracy and trains AI models using wearable and user input data. The app takes in consideration of end users by involving them in research, using inclusive language and visuals, listing diverse symptoms, securing transparent consent, and integrating ongoing feedback.

Utility of expected outcomes and expected benefits for the users/beneficiaries

FlowSync ensures the research results will be implemented in real-life situations by offering an app that addresses unmet needs of women with PMDD, PCOS, and PMS. The app will be tested in real-world settings like home, work, and school to ensure notifications and logging features work within daily routines. The wearable will be tested during exercise, sleep to collect HRV and skin conductance data. Health professionals will also be consulted to check the app's accuracy. To increase engagement, FlowSync will include a user-friendly interface, clear explanations,

tutorials, and daily check-in notifications. The main focus is to provide personalized lifestyle plans to manage hormone imbalances, along with accurate cycle predictions and stress tracking.

References

1. Hirsch, M. D., & Hirsch, M. D. (2020). Physiology, Ovulation And Basal Body Temperature. In *StatPearls*. StatPearls Publishing
2. Schmalenberger, K. M., Eisenlohr-Moul, T. A., & Jarczok, M. N. (2019). Menstrual Cycle Changes in Vagally-Mediated Heart Rate Variability: A Meta-Analysis. *Psychoneuroendocrinology*, 100, 216–226.
3. Girdler, S. S., & Light, K. C. (1990). Skin conductance and heart rate in women with premenstrual syndrome. *Psychophysiology*, 27(5), 577–586
4. Rego, R. C. B. (2023). Predictive Modeling of Menstrual Cycle Length: A Time Series Forecasting Approach. *arXiv preprint arXiv:2308.07927*
5. Urteaga, I., Bertin, T., Hardy, T. M., Albers, D. J., & Elhadad, N. (2019). Multi-Task Gaussian Processes and Dilated Convolutional Networks for Reconstruction of Reproductive Hormonal Dynamics. *arXiv preprint arXiv:1908.10226*